Category Theory for Mathematical Physics — Exercise Sheet 14

Topological quantum field theory: a synthesis

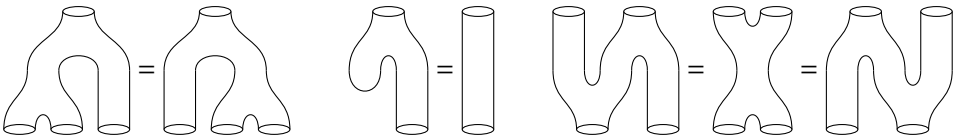
Variant: Solutions

Conventions as in Lecture 14: time runs upward; a Frobenius algebra is $(A, \mu, \iota, \Delta, \varepsilon_A)$, drawn with the pieces of § 14.2 (the pants μ and Δ, the discs ι and ε_A); P is the flip of Vect, $w = \mu \circ \Delta \circ \iota$ the handle element, Σ_g the closed surface of genus $g \geq 0$, and $Z(\Sigma_g) = \varepsilon_A(w^g)$. Groups are multiplicative, with identity 1, and $[a, b] = aba^{-1}b^{-1}$. In this sheet $\beta = \varepsilon_A \circ \mu$ and $\gamma = \Delta \circ \iota$. In Exercise 4, n_0, n_1, n_2 count vertices, edges and faces, and Z_1, B_1 are spaces of chains (not partition functions or centres); in the starred exercise $N(x)$ is a count.

Exercise 1 (The Frobenius axioms as surfaces). (i) Draw as surfaces, glued from the pieces of Lecture 14, the associativity of μ, the unit law $\mu \circ (\iota \otimes \mathrm{id}) = \mathrm{id}$ and the Frobenius relation Equation (14.3). Which pictures give coassociativity and the counit law? Which diffeomorphism of the pair of pants gives commutativity?

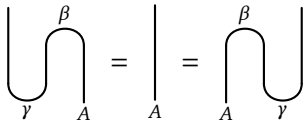
(ii) Show that $(\mathrm{id} \otimes \beta) \circ (\gamma \otimes \mathrm{id}) = \mathrm{id}_A = (\beta \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \gamma)$, draw it as in Equation (14.1), and deduce that β is a nondegenerate pairing with $\beta(xy, z) = \beta(x, yz)$.

Solution. (i) Associativity, $\mu \circ (\mu \otimes \mathrm{id}) = \mu \circ (\mathrm{id} \otimes \mu)$, the unit law and the Frobenius relation are



In each equation both sides are connected surfaces of genus 0 with the same incoming and outgoing circles: a sphere with four holes (three in, one out; or two in, two out), and for the unit law a cylinder, since the disc closes one leg of the pants. Such surfaces are diffeomorphic relative to their boundary (the classification of surfaces, cited: Kock §1.4), so the equations hold in Bord_2, and Z turns them into the axioms. Turning every picture upside down exchanges μ with Δ and ι with ε_A: coassociativity and the counit law. Commutativity, $\mu \circ P = \mu$, is the pants with its two legs exchanged; a half-turn of the legs about the vertical axis, damped to the identity in a collar of the waist, carries this surface to the pants itself, fixing the outgoing circle pointwise and exchanging the incoming ones.

(ii) $(\mathrm{id} \otimes \beta) \circ (\gamma \otimes \mathrm{id}) = (\mathrm{id} \otimes \varepsilon_A)(\mathrm{id} \otimes \mu)(\Delta \otimes \mathrm{id})(\iota \otimes \mathrm{id}) = (\mathrm{id} \otimes \varepsilon_A) \circ \Delta \circ \mu \circ (\iota \otimes \mathrm{id})$ by the Frobenius relation, and $\mu \circ (\iota \otimes \mathrm{id}) = \mathrm{id}$, $(\mathrm{id} \otimes \varepsilon_A) \circ \Delta = \mathrm{id}$ give id_A; the other side uses $(\mu \otimes \mathrm{id})(\mathrm{id} \otimes \Delta) = \Delta \circ \mu$ the same way. In pictures, with the cap β and the cup γ:



As surfaces this is the bent cylinder straightened. Write $\gamma = \sum_i e_i \otimes e^i$. If $\beta(v, z) = 0$ for all z, the right snake gives $v = \sum_i \beta(v, e_i)e^i = 0$; likewise on the other side. So β is nondegenerate, and $\beta(xy, z) = \varepsilon_A(xyz) = \beta(x, yz)$ by associativity.

Exercise 2 (The handle element). (i) Let $A = \mathbb{C}[x]/(x^2)$ with $\varepsilon_A(1) = 0$, $\varepsilon_A(x) = 1$. Show that β is nondegenerate, find $\Delta(1)$ and $\Delta(x)$ from $\Delta(a) = \sum_i ae_i \otimes e^i$, where $\{e_i\}$ and $\{e^i\}$ are dual bases, $\beta(e_i, e^j) = \delta_{ij}$, and compute w and $Z(\Sigma_g)$ for every g.

(ii) For any commutative Frobenius algebra show that $Z(T^2) = \varepsilon_A(w) = \dim A$: the case $\Sigma = S^1$ of Theorem 14.2(ii).

(iii) Let $A = \mathbb{C}[\mathbb{Z}_2]$ with $\varepsilon_A(g) = \delta_{g,1}$. Compute $Z(\Sigma_g)$. What does the counit $\varepsilon_A(f) = f(1)/|G|$ of Theorem 14.6 give instead?

Solution. (i) $\beta(1,1) = 0$, $\beta(1,x) = \beta(x,1) = 1$, $\beta(x,x) = \varepsilon_A(x^2) = 0$: the matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is invertible, and the dual basis of $e_1 = 1$, $e_2 = x$ is $e^1 = x$, $e^2 = 1$. So $\gamma = 1 \otimes x + x \otimes 1$, $\Delta(1) = 1 \otimes x + x \otimes 1$ and $\Delta(x) = x \otimes x + x^2 \otimes 1 = x \otimes x$. The algebra is commutative, so it is a commutative Frobenius algebra (Definition 14.3, with Δ determined by β). Then $w = \mu(\Delta(1)) = 2x$, $w^0 = 1$ and $w^g = 2^g x^g = 0$ for $g \geq 2$: $Z(S^2) = \varepsilon_A(1) = 0$, $Z(T^2) = \varepsilon_A(2x) = 2$, and $Z(\Sigma_g) = 0$ for $g \geq 2$. This algebra is not semisimple: x is nilpotent, and $\mathbb{C}^n$ has no nonzero nilpotents. Its numbers agree up to genus 2 with those of $\mathbb{C}^2$ with $\lambda_1 = \lambda$, $\lambda_2 = -\lambda$ (Example 14.5: $0, 2, 0$), but that theory gives $Z(\Sigma_3) = 2\lambda^{-2} \neq 0$.

(ii) Write $\gamma = \Delta(\iota(1)) = \sum_i e_i \otimes f_i$ in a basis $\{e_i\}$. The snake of Exercise 1(ii) at e_j gives $e_j = \sum_i e_i \beta(f_i, e_j)$, so $\beta(f_i, e_j) = \delta_{ij}$: $f_i = e^i$, the dual basis. Then $\varepsilon_A(w) = \varepsilon_A\big(\sum_i e_i e^i\big) = \sum_i \beta(e_i, e^i) = \sum_i 1 = \dim A$. As a surface, w followed by ε_A is the torus, $S^1 \times S^1$, and the computation is the trace of id_A of Equation (14.2).

(iii) $e_\pm = (1 \pm s)/2$ are orthogonal idempotents with $e_+ + e_- = 1$ and $\varepsilon_A(e_\pm) = \frac{1}{2}$, so by Example 14.5 $Z(\Sigma_g) = 2 \cdot (\frac{1}{2})^{1-g} = 2^g$. For abelian G the class functions with convolution are $\mathbb{C}[G]$ ($f \mapsto \sum_g f(g)\, g$), and $f(1)/|G|$ is $\varepsilon_A/2$; scaling ε_A by $\frac{1}{2}$ multiplies $Z(\Sigma_g)$ by $(\frac{1}{2})^{1-g}$, giving $2^g \cdot 2^{g-1} = 2^{2g-1}$, the Dijkgraaf–Witten value for $\mathbb{Z}_2$ (Exercise 3).

Exercise 4 (Dijkgraaf–Witten for $\mathbb{Z}_2$ and S_3). Use

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \dots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] \rangle.$$

(i) For $G = \mathbb{Z}_2$, count $\#\operatorname{Hom}(\pi_1\Sigma_g, G)$, and check both expressions of Theorem 14.6.

(ii) For $G = S_3$, compute $Z(\Sigma_g)$ for $g = 0, 1, 2$ from the character sum, and $\#\operatorname{Hom}(\pi_1\Sigma_2, S_3)$. Check $g = 0$ and $g = 1$ against the count of homomorphisms directly.

Solution. (i) $\mathbb{Z}_2$ is abelian, so the relation holds for every choice of a_i, b_i: $\#\,\mathrm{Hom} = 2^{2g}$ and $\#\,\mathrm{Hom}/|G| = 2^{2g-1}$. The characters are two, both of dimension 1: $\sum_\chi (2/1)^{2g-2} = 2 \cdot 2^{2g-2} = 2^{2g-1}$. They agree for every g.

(ii) $\dim \chi = 1, 1, 2$ and $|G| = 6$: $Z(\Sigma_g) = 6^{2g-2} + 6^{2g-2} + 3^{2g-2}$, so $Z(S^2) = \frac{1}{36} + \frac{1}{36} + \frac{1}{9} = \frac{1}{6}$, $Z(T^2) = 3$ and $Z(\Sigma_2) = 36 + 36 + 9 = 81$, with $\#\operatorname{Hom}(\pi_1\Sigma_2, S_3) = 81 \cdot 6 = 486$. At $g = 0$ the group is trivial, one homomorphism, and $1/|G| = \frac{1}{6}$. At $g = 1$ a homomorphism is a commuting pair (a, b): for each a there are $|C(a)|$ choices of b, so the count is $1 \cdot 6 + 3 \cdot 2 + 2 \cdot 3 = 18 = |G| \cdot 3$ (each class contributes $|K| \cdot |G| / |K|$), and $Z(T^2) = 3$, the number of classes, which is $\dim A$ as Exercise 2(ii) requires.

Exercise 4 (The Levin–Wen count for $\mathrm{Vec}_{\mathbb{Z}_2}$). Let Γ be a connected trivalent graph in Σ_g whose faces are discs, with n_0 vertices, n_1 edges and n_2 faces, so that $n_0 - n_1 + n_2 = 2 - 2g$. The string net of $\mathrm{Vec}_{\mathbb{Z}_2}$ puts a qubit, a label 0 or 1, on each edge; $Q_v = 1$ when the labels at v add up to 0 mod 2, and $B_p = \frac{1}{2}(1 + X_p)$, where X_p adds ∂p, the boundary of the face p mod 2, to the labels (an edge along which ∂p runs twice is not flipped) (§ 14.4, with every $d = 1$ and every F-symbol 1).

(i) Let Z_1 be the closed $\mathbb{Z}_2$-chains (the label sets with $Q_v = 1$ at every v) and B_1 the sums of face boundaries. Show that the ground states are the functions on Z_1 that are constant on the cosets of B_1.

(ii) Show that $\dim Z_1 = n_1 - n_0 + 1$ and $\dim B_1 = n_2 - 1$ over $\mathbb{Z}_2$, and deduce that there are 2^{2g} ground states.

(iii) Compare the torus with Theorem 14.8(ii)–(iii) and with *Quantum Computing*. How many torus states does Theorem 14.8 give for Vec_{S_3}, and why is that not the $Z(T^2) = 3$ of Exercise 3?

Solution. (i) The Q_v and B_p commute (∂p is a closed chain, so X_p changes an even number of labels at each vertex), and a state with every $Q_v = 1$ and every $B_p = 1$ exists (the one below), so the ground states are exactly those. $Q_v = 1$ for all v means the state is supported on label sets with an even number of 1's at each vertex: $\psi = \sum_{z \in Z_1} \psi(z)\,|z\rangle$. $B_p\psi = \psi$ means $X_p\psi = \psi$, that is $\psi(z + \partial p) = \psi(z)$. So ψ is constant on each coset $z + B_1$, and every such function is a ground state.

(ii) The boundary map $\partial\colon \mathbb{Z}_2^{n_1} \to \mathbb{Z}_2^{n_0}$ sends an edge to its two ends; its image consists of the sets of vertices of even size (as Γ is connected, any two vertices are joined by a path, whose boundary is the pair), of dimension $n_0 - 1$, so $\dim Z_1 = \dim\ker\partial = n_1 - n_0 + 1$. The face map $\mathbb{Z}_2^{n_2} \to \mathbb{Z}_2^{n_1}$ sends a face to its boundary; every edge has two face-sides (counted with multiplicity), and the faces are connected through edges, so a set of faces with no boundary is empty or everything: the kernel has dimension 1 and $\dim B_1 = n_2 - 1$. Each face boundary is closed, so $B_1 \subseteq Z_1$, and the number of ground states is $|Z_1/B_1| = 2^{(n_1-n_0+1)-(n_2-1)} = 2^{2-(n_0-n_1+n_2)} = 2^{2g}$: one on the sphere, four on the torus.

(iii) On the torus, 4 is $\dim Z_{\mathrm{TV},\mathrm{Vec}_{\mathbb{Z}_2}}(T^2) = \dim \mathrm{RT}_{\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})}(T^2) = \operatorname{rank}\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, by Theorem 14.8(ii)–(iii), and the four ground states of the toric code (*Quantum Computing* Thm. 14.2): the string net of $\mathrm{Vec}_{\mathbb{Z}_2}$ is the toric code on the dual lattice. For Vec_{S_3} the same gives $\operatorname{rank}\mathcal{Z}(\mathrm{Vec}_{S_3}) = 8$ (Sheet 12, Exercise 3). The 3 of Exercise 3 is the partition function of the torus in the two-dimensional theory, $\dim Z(S^1)$, the number of classes; the 8 is the dimension of the state space on the torus in the three-dimensional theory. Different theories, in different dimensions.

Exercise 5 (Turaev–Viro, Reshetikhin–Turaev and the anomaly). (i) Give $\dim Z_{\mathrm{TV},C}(T^2)$ for $C = \mathrm{Vec}_{\mathbb{Z}_2}$, Vec_{S_3}, Fibonacci and Ising.

(ii) For Fibonacci, Ising, $\mathrm{SU}(2)_2$, the toric code and $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$, find c mod 8 (Lecture 10) and say whether the gluing phase $e^{2\pi i c/8}$ of RT is 1.

(iii) Why is no Turaev–Viro theory equal to $\mathrm{RT}_{\mathrm{Fib}}$?

Solution. (i) By Theorem 14.8(ii)–(iii), $\dim Z_{\mathrm{TV},C}(T^2) = \operatorname{rank}\mathcal{Z}(C)$: 4 for $\mathrm{Vec}_{\mathbb{Z}_2}$ (Theorem 12.8), 8 for Vec_{S_3} (Sheet 12, Exercise 3), and, as Fibonacci and Ising are modular, $\mathcal{Z}(C) \simeq C \boxtimes C^{\mathrm{rev}}$ (Theorem 12.10) of rank $2^2 = 4$ and $3^2 = 9$.

(ii) From the Gauss sums of Lecture 10: Fibonacci $c = \frac{14}{5}$, Ising $\frac{1}{2}$, $\mathrm{SU}(2)_2$ $\frac{3}{2}$ ($3k/(k+2)$ at $k = 2$), the toric code 0, and $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ 0 (its Gauss sum is $|p_+|^2 > 0$, Sheet 12, Exercise 5). The phase $e^{2\pi i c/8}$ is 1 only for the last two, the centres; the other three give anomalous RT, and with Bakalov and Kirillov's negative root allowed they would still need $c \equiv 0 \pmod 4$.

(iii) By Theorem 14.8(ii) every Turaev–Viro theory is RT of a centre, and a centre has $c = 0$ (Lecture 12). $\mathrm{RT}_{\mathrm{Fib}}$ has $c = \frac{14}{5} \neq 0$, and the central charge, read off from S and T on the torus, is an invariant of the theory. The closest is $Z_{\mathrm{TV},\mathrm{Fib}} = \mathrm{RT}_{\mathrm{Fib}\boxtimes\mathrm{Fib}^{\mathrm{rev}}}$, the doubled Fibonacci.

Exercise 6 ⋆. *Mednykh's formula.* Let G be a finite group, K its conjugacy classes, and for $x \in G$ let

$$\mathcal{N}(x) = \#\{(a,b) \in G^2 : [a,b] = x\}.$$

You may use, for the irreducible characters, $1_K = \frac{|K|}{|G|}\sum_\chi \overline{\chi(K)}\,\chi$ (the indicator function of the class), the relation $\sum_K |K|\ |\chi(K)|^2 = |G|$, and $\chi * \psi = \delta_{\chi\psi}\,(|G|/\chi(1))\,\chi$ for the convolution of Theorem 14.6.

(i) Show that $\mathcal{N} = \sum_K \frac{|G|}{|K|}\, 1_{K^{-1}} * 1_K$.

(ii) Deduce that $\mathcal{N}(x) = |G| \sum_\chi \chi(x)/\chi(1)$.

(iii) Show that $\#\operatorname{Hom}(\pi_1\Sigma_g, G) = \mathcal{N}^{*g}(1)$, and conclude that $\#\operatorname{Hom}(\pi_1\Sigma_g, G)/|G| = \sum_\chi (|G|/\chi(1))^{2g-2}$.

Solution. (i) Fix a. Then $[a,b] = x$ says $ba^{-1}b^{-1} = a^{-1}x$. As b runs over G, $ba^{-1}b^{-1}$ runs over the class K_a of a^{-1}, each element $\left|C(a^{-1})\right| = |G|/|K_a|$ times. So $\#\{b : [a,b] = x\} = \frac{|G|}{|K_a|} 1_{K_a}(a^{-1}x)$. Group the a by the class K of a^{-1}, that is $a \in K^{-1}$:

$$\mathcal{N}(x) = \sum_K \frac{|G|}{|K|} \sum_a 1_{K^{-1}}(a)\, 1_K(a^{-1}x) = \sum_K \frac{|G|}{|K|}\, (1_{K^{-1}} * 1_K)(x),$$

since $(f * h)(x) = \sum_{yz=x} f(y)h(z)$ with $y = a$, $z = a^{-1}x$.

(ii) $\chi(K^{-1}) = \overline{\chi(K)}$, so $1_{K^{-1}} = \frac{|K|}{|G|} \sum_\chi \chi(K)\chi$ and $1_K = \frac{|K|}{|G|} \sum_\psi \overline{\psi(K)}\psi$. By the convolution identity only $\psi = \chi$ survives:

$$1_{K^{-1}} * 1_K = \frac{|K|^2}{|G|^2} \sum_\chi \chi(K)\overline{\chi(K)}\, \frac{|G|}{\chi(1)}\, \chi = \frac{|K|^2}{|G|} \sum_\chi |\chi(K)|^2\, \frac{\chi}{\chi(1)}.$$

Then $\mathcal{N} = \sum_K |K| \sum_\chi |\chi(K)|^2 \chi/\chi(1) = \sum_\chi \frac{\chi}{\chi(1)} \sum_K |K|\, |\chi(K)|^2 = |G| \sum_\chi \chi/\chi(1)$.

(iii) A homomorphism is a choice of a_i, b_i with $[a_1,b_1]\cdots[a_g,b_g] = 1$. Sorting by $x_i = [a_i,b_i]$, the number is $\sum_{x_1\cdots x_g=1} \mathcal{N}(x_1)\cdots\mathcal{N}(x_g) = \mathcal{N}^{*g}(1)$. At $g = 0$ both sides are 1 (one homomorphism; $\mathcal{N}^{*0} = \delta_1$), and the formula agrees, as $\sum_\chi \chi(1)^2 = |G|$ (the given 1_K at $K = \{1\}$, evaluated at 1). For $g \geq 1$, by (ii) and the convolution identity, $\chi^{*g} = (|G|/\chi(1))^{g-1}\chi$, so

$$\mathcal{N}^{*g}(1) = |G|^g \sum_\chi \frac{1}{\chi(1)^g} \Big(\frac{|G|}{\chi(1)}\Big)^{g-1} \chi(1) = |G|^{2g-1} \sum_\chi \chi(1)^{2-2g},$$

and dividing by $|G|$ gives $\sum_\chi (|G|/\chi(1))^{2g-2}$: the first expression of Theorem 14.6 equals the second, and Mednykh's theorem is proved. For $\mathbb{Z}_2$ and S_3 it gives Exercise 3's values ($\mathcal{N}(1) = 18$ and $\mathcal{N}^{*2}(1) = 486$ for S_3; the script also checks $\mathcal{N}$ itself on S_3, and the formula by brute force for three groups of orders 8 and 12).